Explain the 6 stages of the Data Science Life Cycle using Netflix's movie recommendation system

For each stage, briefly explain:
What the stage means
How Netflix uses it
Main activities performed
Expected outcome
Cover the following stages:
Business Understanding
Data Collection
Data Preparation
Exploratory Data Analysis (EDA)
Model Building & Evaluation
Deployment & Monitoring

Scenario:
Netflix wants to recommend movies and TV shows that users are most likely to watch, increasing watch time and customer satisfaction.

Requirements:
Keep the explanation simple and practical.
Avoid technical jargon where possible.
Use short paragraphs or bullet points.
Include easy-to-understand examples from Netflix.
Keep the entire response within 2–3 pages (approximately 800–1000 words).
End with a short conclusion summarizing how the six stages help Netflix improve recommendations.

Netflix uses the data science life cycle to turn user behavior into better movie and show recommendations. In simple terms, it learns what people like, tests ideas, and keeps improving the suggestions it shows on the homepage.

1) Business Understanding

This stage is about defining the problem clearly. Netflix's goal is to keep users watching longer and make them happier with the content they see.

- What it means: Understand the business goal before working with data.
- How Netflix uses it: Netflix wants to recommend titles that a user is likely to watch and enjoy.
- Main activities: Identify the problem, define success, and decide what the recommendation system should achieve.
- Expected outcome: A clear goal, such as “show the most relevant movies and TV shows to each user.”

Example: If a user often watches action movies, Netflix wants the homepage to show more action titles first.

2) Data Collection

This stage means gathering useful information from different sources. Netflix collects data from user actions and from details about the shows themselves.

- What it means: Collect raw information that may help solve the problem.
- How Netflix uses it: Netflix looks at viewing history, ratings, how long a title was watched, the time of day, language preferences, device type, and title details like genre, actors, and release year.
- Main activities: Record clicks, searches, watches, ratings, and content information.
- Expected outcome: A large set of data that shows what users do and what each title is like.

Example: If many users who watch crime dramas also watch a certain thriller, that pattern becomes useful data.

3) Data Preparation

Raw data is often messy, incomplete, or repeated, so it must be cleaned and organized. Netflix prepares the data so it can be used for prediction and analysis.

- What it means: Clean, sort, and format the data so it is ready to use.
- How Netflix uses it: Netflix organizes watch history, user preferences, and title information into a form that the recommendation system can understand.
- Main activities: Remove errors, fill missing values, combine data sources, and create helpful features such as “preferred genre” or “usual watch time.”
- Expected outcome: A clean and useful dataset that is ready for analysis and model training.

Example: If a user watched only 5 minutes of a movie, Netflix may treat that differently from a movie they finished.

4) Exploratory Data Analysis

EDA means exploring the data to find patterns, trends, and surprises before building a model. It helps Netflix understand what the data is telling it.

- What it means: Study the data to discover patterns and relationships.
- How Netflix uses it: Netflix can look for links between viewing habits, genres, time of day, language, and what users enjoy most.
- Main activities: Create charts, compare groups, find common watching patterns, and check which features matter most.
- Expected outcome: Better understanding of user behavior and content trends.

Example: Netflix may notice that a user watches comedies more often on weekends and short episodes on weekdays.

5) Model Building & Evaluation

This stage is where Netflix creates the recommendation system and checks how well it works. The system predicts what a user is most likely to enjoy based on past behavior and similar users.

- What it means: Build a model that makes predictions, then test how accurate it is.
- How Netflix uses it: Netflix updates its algorithms using signals from watching, finishing titles, and giving thumbs up or other feedback. Some reports also mention A/B testing to compare different recommendation versions.
- Main activities: Train the model, test it on data, compare different versions, and measure accuracy or usefulness.
- Expected outcome: A recommendation model that can suggest relevant titles more effectively than a random list.

Example: If a user watches several superhero films, the model may suggest more action or fantasy movies.

6) Deployment & Monitoring

Deployment means putting the model into real use, and monitoring means watching how it performs over time. Netflix does not stop after building the system; it keeps learning from new user behavior.

- What it means: Use the model in the real product and keep checking it.
- How Netflix uses it: Netflix updates recommendations continuously based on fresh user actions such as what people start, finish, or rate.
- Main activities: Launch the model, collect live feedback, track performance, and retrain the system when needed.

- Expected outcome: Recommendations stay current, useful, and aligned with changing user tastes.

Example: If a user starts watching more documentaries, Netflix should adjust and recommend more documentaries soon.

Conclusion

All six stages work together to help Netflix recommend better content. By understanding the goal, collecting the right data, cleaning it, studying patterns, building a good model, and updating it regularly, Netflix can improve watch time and user satisfaction.